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Research

The reasons for turnover of information systems personnel *

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The study examines the effects of the individual characteristics, job type, role stressors, boundary spanning activities, career outcomes, and job characteristics on the turnover propensity of 464 information systems personnel. Results show that age, organizational level, organizational tenure, job tenure, and number of years in the computer field are negatively correlated with the intention to leave the organization. Education was found to be positively correlated with turnover intentions, and while project leaders are more likely to leave the organization, IS managers are less likely. Results also show that both role stressors (role ambiguity and role conflict) and boundary spanning activities are positively correlated with turnover intentions, and that job involvement, career plateau, promotability, salary, organizational commitment, job satisfaction, satisfaction with progress, promotion, pay, status, and projects are negatively correlated while career opportunity is positively correlated with turnover intentions. Finally, all job characteristics are negatively correlated with turnover intentions. Implications of the results for practice and research are offered.

Keywords: Turnover correlates; Retention decisions; Information systems personnel; Organizational commitment; Career decisions; Satisfaction

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1. Introduction

Managing the human resources within IS departments is being closely scrutinized in many organizations. One reason is that the cost of IS personnel is high; it consumed an average of 43% of the entire IS departmental budgets in 1988

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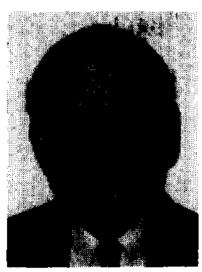

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[10]. IS executives feel that, if anything, this figure is expected to continue to escalate in the coming years. Increased productivity through higher levels of motivation and greater retention rates of these information systems employees could help contain costs. It is not surprising that the initial steps of hiring and training IS personnel are now being more carefully studied. Factors such as reported chronic shortages, the high costs of recruiting and training these specialized employees, turnover rates averaging about 20% per year, and a demand-side growth rate that is increasing by 15% per year all point to a need for more effective human resource management of IS personnel. Employee turnover, although not the only contributor to escalating costs, does play a significant role. Beyond the obvious implication of direct costs associated with it, there are secondary ramifications to be considered. For example, the constant replacement of personnel could produce a disproportionate ratio of employees in the 'introductory' stages of the learning curve instead of the more productive 'maturity' stages. And, those employees remaining within the IS department could become somewhat demoralized as a result of an organizational malaise that occurs when turnover becomes excessive. Depending on the physical isolation of the IS department, there could even be spin-off morale problems in other departments. Thus, there is a growing need to control the increasing volume of voluntary turnover actions of IS employees.

As is typical in most management approaches for creating change in organizational behavior, the starting point is to understand the forces behind the existing behavior-turnover and to define and assess those aspects of jobs being conducted by IS personnel that are within the control of IS management and are possibly contributing to turnover. To the extent that appropriate administrative actions could be designed to address these problem areas, personnel turnover could be decreased.

This paper examines turnover propensity of IS employees and factors affecting it. The objective is to determine the underlying reasons that affect the intention of employees to stay in or leave the organization and to present viable options to manage this growing problem. To achieve this, a range of individual, work, and organizational characteristics and conditions are discussed along with their resultant impact on an individual's decision.

Literature within the organizational behavior and IS areas identify different factors affecting turnover intentions of both technical and non-technical employees. Satisfaction with job, promotion, status, progress, and types of projects were found to affect the individual intentions to leave the organization. In addition, high levels of stress, common in the IS environment due to rapid technological change, are reported to create role stressors that also impact on turnover decisions. While it is apparent that physiological and social needs satisfaction levels affect the turnover decision, it has been reported that job factors more closely associated with ego needs, such as autonomy, respect, advancement, recognition, challenge, and achievement have greater influence on turnover. Studies find that, for IS personnel, issues such as loyalty and commitment to the organization are directly related to the voluntary turnover decision. More recently, boundary spanning activities requiring individuals to cross departmental, functional, and organizational lines as part of their job is considered one of the major variables affecting turnover decisions. The nature of the IS function in most organizations makes these personnel vulnerable to this condition.

In summary, the literature suggests that the organizational tenure of IS personnel is related to a series of potential factors that can be separated into five broad categories: First, the demographic characteristics, such as age, gender, length of time in current job and with the organization; second, sources of intrinsic satisfaction in the job and career, such as promotability and involvement; third, task characteristics related to the specific job; fourth, organizational attitudes, such as loyalty and commitment; and finally, boundary spanning activities. Based on these categories a study was designed to analyze the correlates and their impact on voluntary turnover decisions for IS employees.

2. The study design

2.1. Sample and procedure

A questionnaire was distributed to 2548 members of the Association for Computing Machinery

(ACM) in Pennsylvania, Delaware, and Southern New Jersey. The sample was chosen because members of ACM represent a wide variety of jobs and organizational settings. Five hundred and seventeen questionnaires (response rate of 20.3%) were returned to the researchers. This response rate appears to be consistent with other mail surveys Fifty three questionnaires were elimi-

Table 1
Demographic characteristics of the sample ($N = 464$).

	Our sample (%)	ACM population (%) [b]
Gender		
Male (1) [a]	80.3	85.0
Female (2)	19.7	15.0
Education		
Some high school or less (1)	0.0	–
High school (2)	0.2	–
Some college (3)	5.2	–
Bachelor's degree (4)	21.3	–
Some graduate school (5)	21.9	–
Graduate degree (6)	51.4	56.0
Had at least a bachelor's degree	94.6	92.0
Marital status		
Unmarried (1)	31.8	–
Married (2)	68.2	–
Organizational level		
Professional (1)	53.9	–
Management (2)	46.1	–
Salary categories		
Below $25000 (1)	4.2	–
25000–34999 (2)	16.4	–
35000–44999 (3)	24.4	–
45000–54999 (4)	20.5	–
55000–64999 (5)	15.3	–
65000 or above (6)	19.2	–
Size of computer department (number of employees)		
1–5 (1)	12.7	–
6–10 (2)	9.8	–
11–20 (3)	11.1	–
21–30 (4)	7.4	–
Above 30 (5)	59.0	–
Age		
Mean = 37.76 S.D. = 9.46 Range = 22–67		
55 Years old or younger	93.2	94.0
Organizational tenure (in years)		
Mean = 6.27 S.D. = 6.18 Range = 1–36		
Job Tenure (in Years)		
Mean = 3.67 S.D. = 3.69 Range = 0.5–28		
Number of Years in MIS Field		
Mean = 11.96 S.D. = 8.08 Range = 1–36		
Nine years or less	49.5	48.0
Fourteen years or less	67.7	68.0
More than 25 years	10.0	9.0

Position (% of total):

System programmers	7.5%	Systems analysts	13.6%
Project leaders	12.9%	IS managers	20.0%
Applications programmers	17.0%	Software engineers	18.8%
Consultants	10.0%	Unidentified	0.2%

[a] The numbers in parentheses represent the coding for the value of the specific variable.
[b] The percentages are based on the work of Fletcher et al. [5]. Other figures were not reported.

nated due to missing data or subject unsuitability (retired employees and faculty at academic institutions), leaving a final sample of 464 employees (response rate of 18.2%). The participants in this study held a variety of positions within the computer field reflecting considerable heterogeneity within the MIS profession. *Table 1* presents a summary of the demographic characteristics of the sample.

The sample was validated by determining the representation of the respondents to the population of ACM membership (see Table 1). Fletcher et al. [5] reported that 85% of ACM members were men, and 94% of the members were 55 years old or younger, whereas 80.3% of our sample were men and 93.2% reported to be 55 years old or younger. With regard to education, Fletcher et al. [5] reported that 56% of ACM members held a master's degree (51% in our sample), and 92% had at least a Bachelor's degree (compared to 94% in our sample). We found that 49.5% of the participants had fewer than 9 years of computer experience (versus 48% in the ACM report), 67.7% had fewer than 14 years of computer experience (versus 68% in the ACM report), and 10% of the sample respondents had more than 25 years of experience in computers (versus 9% in the ACM report). These data provide substantial evidence that the demographic characteristics of our sample are quite similar to those of the total ACM population.

2.2. Measures

Gender was assessed by: 1 = male, 2 = female. Responses to *marital status* were also coded as a binary variable (1 = not married; 2 = married). Respondents indicated their *job titles* in an open-ended item. *Education* was the highest of six levels from (1) some high school to (6) graduate or professional degree. *Age* was measured in years. *Organizational level* consisted of two levels: (1) professionals; or (2) supervisors and managers. *Tenure* in the current job and organization was measured in number of years. Participants were also asked to indicate how many years they had been in the *MIS field*. The demographic items were included as background information.

Role stressors. Two types of role stressors were measured: Role ambiguity and role conflict. Role ambiguity is the extent to which an employee is unclear about his or her job duties, performance standards, or level of job performance. Role conflict refers to incompatible expectations associated with a work role. It often happens that two or more demands occur simultaneously, and then complying with one makes it difficult to comply with the other.

Role ambiguity and *role conflict* were measured with six and eight items respectively from scales developed by Rizzo et al. [14]. Each scale was scored using a 5-point response mode from (1) very false to (5) very true. The responses to the role conflict and role ambiguity items were coded so that the greater the score, the greater the perceived ambiguity or conflict. The six role ambiguity items (e.g., "I know exactly what is expected of me"; "I receive a clear explanation of what has to be done") were averaged to obtain an overall index of role ambiguity. Similarly, the eight role conflict items (e.g., "I receive incompatible requests from two or more people"; "I have to do things in a certain way even though they should be done differently") were averaged to create an overall role conflict score. The internal consistency reliability of role ambiguity and role conflict was 0.81 and 0.87, respectively.

Perceived job characteristics. Thirteen items were used to assess the extent to which employees' current jobs provided opportunities and rewards. The response options were anchored on a five-point scale ranging from (1) not at all to (5) to a great extent. Eight items were adapted from Allen and Katz [1] and five more items were added to measure different aspects of the job. A factor analysis (with varimax rotation) produced two factors with eigenvalues ≥ 1.0 that accounted for 58.3% of the total variance. Factor 1, labeled *task-based rewards*, included the following items: Pursue your ideas, build a professional reputation, work with competent colleagues, work on technically challenging tasks, work on professionally important projects, have freedom to be creative and original, and be highly respected by your peers. Responses to the seven items of this factor were averaged to create a scale reflecting the task-based component. The internal consistency reliability of the scale was 0.88.

Factor 2, labeled *organization-based rewards*, included work on organizationally important projects, work on projects leading to advancement, be highly respected by top management, receive substantial annual salary increases, have a great deal of power and influence on the job, and receive a promotion within the next year or two. Responses to the six items were averaged to create a scale reflecting the organizational component of the job. The internal consistency reliability of the scale was 0.84.

Boundary spanning activities, defined as those by which employees interact and communicate with others outside their departments, including formal and informal written and oral communications, were measured by modifying the ten-item scale developed by Miles and Perreault [12] and used by Baroudi [2]. Each item (e.g., "assist other units in determining appropriate uses of information technology"; "recommend new applications of information technology to top management") was scored on a five-point scale ranging from (1) not part of my job to (5) a very significant part of my job. The internal consistency reliability of the scale was 0.91.

Job involvement refers to the degree to which an employee identifies psychologically with his or her present job and the extent to which the job situation is central to the employee's self-identity. Job involvement was measured by an eight-item scale adapted from Kanungo [9]. Individuals indicated their agreement or disagreement with each statement (e.g., "I consider my job to be very central to my existence"; "most of my interests are centered around my job") on a five point Likert-type scale ranging from (1) strongly disagree to (5) strongly agree. Responses to the eight items were averaged to create a job involvement score. The internal consistency reliability of the scale was 0.85.

Advancement prospects were assessed in two ways. A *promotability* assessment was measured with the item: "How would you rate your chances for promotion to a higher position sometime during your career with the company?" Responses were made on a three-point scale with the following anchors: (1) slight chance for promotion; (2) good chance for promotion; and (3) very good chance for promotion.

Advancement prospects were also measured by a *plateau* assessment, which determined whether the employees had reached plateaus in their careers. The measure of employee career plateau status was based on tenure in the current job. Current research reports that employees are most likely to leave in their first seven years with the company. Therefore, employees in this study were considered plateaued (coded 2) if their tenure was seven years or more and otherwise coded 1. Although job tenure does not directly assess plateau status, it is generally assumed that extensive length of tenure in one position reflects limited prospects for upward mobility.

The measure of *salary* was based on annual salary in current position. Categories ranged from (1) below $25 000 to (6) $65 000 or above. In order to assess employee feelings about *career opportunities*, two items asked the respondents to indicate their belief about alternative positions outside the organization: "There are excellent opportunities at the present time to find a job in a different organization that is acceptable to me", and "If you search for an alternative job, how likely is it that you could obtain an acceptable job in another organization". The first item had a five-point scale ranging from (1) strongly disagree to (5) strongly agree, and the second item a five-point scale from (1) very unlikely to (5) very likely (alpha = 0.86).

Career satisfaction refers to the emotional reactions of employees to the progress they have made in their careers. This was measured on a five-item scale adapted from prior research [7], with appropriate changes to make the items more relevant to the current sample. Individuals indicated their agreement or disagreement with each statement (e.g., "I am satisfied with the progress have made toward achieving my overall career goals") on a five point Likert-type scale ranging from (1) strongly disagree to (5) strongly agree. Responses were averaged to create a career satisfaction score. The internal consistency reliability of the scale was 0.89.

Job satisfaction refers to the emotional reactions of individuals to their job and its experiences. Job satisfaction was assessed with a three-item scale [8] reflecting overall satisfaction with the job. Each item (e.g., "Generally speaking I am satisfied with my job") required the respondents to indicate their agreement or disagreement on a five point scale ranging from (1) strongly disagree to (5) strongly agree. Responses were averaged to produce a total job satisfaction score. The internal consistency reliability of the scale was 0.77.

Organizational commitment is defined as identification with a particular organization and the desire to maintain membership in it. This was measured by an abbreviated version of the organizational commitment questionnaire (OCQ) developed by Porter et al. [13]. The shorter version represents a more 'pure' measure of the affective dimensions of commitment to the employing organization. Sample items are "I really care about the fate of my organization"; and "I feel very little loyalty to my organization". The response options to the items ranged from (1) strongly disagree to (5) strongly agree. The items were coded such that high scores reflected greater commitment to the organization. The internal consistency reliability of the scale was 0.91.

Intention to stay was measured by a single item: "Given everything you know about the company in which you are employed and the type of work you like to do, how long do you think you will continue to work at this company?" The response options were anchored on a time-linked five-point scale ranging from (1) one year or less to (5) eleven years or more or until retirement. The item was scored such that high scores reflected stronger intentions to stay with the organization.

3. Results and discussion

3.1. Intention to stay with the organization

The median time of intention to stay in the company was 4–6 years; 63.3% of the respondents indicated that they would be staying with the company for less than 6 years. Among this group 16.3% indicated that they would be leaving within a year, 26.5% within 2–3 years and 20.5% within 4–6 years. Results also show that the median length of time in their present organization was only 4 years and in their present job was 2 years.

3.2. Relationships with other factors

(1) *Individual characteristics*. We used demographic factors such as age, gender, organizational tenure, job tenure, number of years in the computer field, education, and organizational level. The relationships between them and turnover propensity are presented in *Table 2*.

Table 2
Relationships between individual characteristics and intention to leave the organization ($N = 464$).

Gender	−0.01
Marital status	−0.09 [a)]
Age	−0.032 [b)]
Organizational level	−0.21 [b)]
Organizational tenure	−0.30 [b)]
Job tenure	−0.19 [b)]
Years in the computer field	−0.27 [b)]
Education	0.15 [b)]

a) $p \leq 0.05$.
b) $p \leq 0.001$.

This shows that age, organizational level, organizational tenure and job tenure are negatively correlated with turnover propensity. On the other hand, education was found to be positively correlated with intention to leave the organization. Results conform to the general finding of correlations between these and intention to leave. There is no evidence that women are either more or less likely to leave the organization than men. On the other hand, unmarried are more likely to leave the organization than married ones. IS employees who are highly educated, unmarried, with few years in both the organization and their position may have less intention to stay with the organization than others. Possibly, career opportunities for older and more veteran employees are limited and they are thus more likely to stay with existing organizations once they reach a career plateau.

(2) *Job type*. The intention to leave the organization as a function of different types of jobs (e.g., systems programmers, applications programmers, software engineers, systems analysts, project leaders, computer managers and consultants) was examined. *Table 3* presents the results using analysis of variance and F-statistics. It shows that there are significant differences in turnover intentions. Computer managers are the least likely to leave the organization, where software engineers and project leaders are most likely to leave.

(3) *Role stressors and boundary spanning activities*. The two role stressors (role ambiguity and role conflict) were found to be positively correlated with intention to leave. *Table 4* shows that the greater the levels of role conflict and ambiguity experienced by employees, the stronger their intention to leave. IS management should take concrete steps to reduce these conditions: e.g.,

Table 3
Relationships between job type and intention to leave the organization ($N = 464$) [a].

Job type	Intention to leave
Systems programmers ($N = 35$)	3.03
Application programmers ($N = 79$)	3.04
Software engineers ($N = 87$)	3.16
Systems analysts ($N = 63$)	3.05
Project leaders ($N = 60$)	3.19
Computer managers ($N = 93$)	2.49
Consultants ($N = 46$)	2.84

[a] The relationship between job type and intention to leave, using analysis of variance, was found to be statistically significant at level 0.05 ($F = 2.25$, $p = 0.039$).

establish clear priorities, develop two way communications systems to insure congruence of expectations, avoid unrealistic deadlines, maintain direct reporting networks, and structure tasks so they are capable of being performed within the desired dimensions of time, cost and quality.

Management may also try to minimize conflicts between IS personnel and end users wherever possible, as well as being supportive of these same personnel when conflict cannot be avoided. Management should arrange co-ordination meetings to keep IS employees up to date and resolve conflicting demands from end users and IS management. They should also provide an environment in which conflicts can be discussed and resolved. Implementation of these suggestions could reduce role conflict, role ambiguity, and role stress, and eventually the likelihood of employees deciding to leave the organization.

Boundary spanning activities for IS personnel were found to be negatively correlated with the intention to leave the organization. This indicates that IS employees view interaction with people in other departments as desirable, perhaps an important part of their career development. Boundary spanning activities may make the job more interesting and IS personnel may also feel that their work has an impact on the organization if it encompasses several departments. This may also provide intangible rewards, such as respect, recognition and advancement for IS personnel; this will reduce their intention to leave the organization. The organization should also encourage having committees of several representatives from different functional areas so IS employees can interact and communicate more effectively.

Table 4
Relationships between role stressors and boundary spanning activities, and intention to leave the organization ($N = 464$).

Role ambiguity	0.21 [a]
Role conflict	0.27 [a]
Boundary spanning activities	−0.19 [a]

[a] $p \leq 0.001$.

Table 5
Relationships between career outcomes and intention to leave the organization ($N = 464$).

Job involvement	−0.17 [b]
Advancement prospects:	
Career plateau status [a]	−0.18 [b]
Promotability	−0.22 [b]
Salary	−0.25 [b]
Career opportunities	0.15 [b]
Organizational commitment	−0.15 [b]
Overall job satisfaction	−0.52 [b]
Overall career satisfaction	−0.31 [b]
Satisfaction with progress	−0.27 [b]
Satisfaction with promotion	−0.23 [b]
Satisfaction with pay	−0.16 [b]
Satisfaction with status	−0.28 [b]
Satisfaction with projects	−0.32 [b]

[a] 1 = not plateaued; 2 = plateaued.
[b] $p \leq 0.001$.

(4) *Career related variables*. Involvement, advancement, salary and career opportunities. The relationship between job involvement and the intention to leave the organization is negative (see *Table 5*). The highly-involved IS personnel are more likely to stay with the organization than the less-involved ones. We can assume that the IS employee who considers his/her job important, and is therefore highly involved in it, would also feel that he/she is an important member of the IS department, involved in what is happening within it, and having some influence and control over the environment. These conditions will lead to more satisfaction in the job and career, and eventually positively impact on the decision to stay. It should be noted that the more an organization encourages its IS employees to be involved, the more they will be satisfied with their jobs and careers and the more likely to stay with the organization. Organizations can encourage involvement by providing IS employees with opportunities to make important job decisions, and incorporating more challenge and interest into jobs and projects.

The career plateau status is negatively correlated with intention to leave the organization. This suggests that, while employees were most likely to leave the organization in their first seven years, after this time period the likelihood of voluntary departure dropped dramatically. It was reported that plateaued employees may become angry, frustrated, bored, stagnant, and less involved and motivated in work [6] ultimately leading to deteriorating performance. At the same time, the absence of further promotional opportunities is not always viewed as a negative factor. Today, many employees may not care for further advancement because of negative consequences, such as geographical relocation; this is important in the era of two career families. Such employees could develop off-the-job interests to compensate for lack of ego satisfaction, and thus their job performance might not suffer. Organizations need to differentiate between two types of plateaued employees: (1) those who are still performing effectively; and (2) "dead-wood" whose job performance is substandard. The effective performers could be kept active by management doing the following: Providing expanded and flexible opportunities for those seeking mobility; increasing utilization of individuals within their current jobs; understanding of the unique problems facing these employees; enhancing training and continuing education programs; and broadening the reward system.

Promotability was also found to be negatively correlated with the intention to leave the organization. This indicates that the more promotable employees are less likely to leave. The obvious caveat here is that management's failure to live up to expectations could have future negative consequences. For those IS employees facing limited advancement opportunities, there is a desire to leave the organization rather than compromise career goals. These employees may change jobs frequently in the hope of finding the promotional opportunities they feel they deserve. Organizations need to find a way to retain their best IS employees and promote those who are most promising. Other studies have noted that IS employees ranked intrinsic qualities of their job, such as sense of achievement and advancement, higher than pay and benefit [3,4]. Accordingly, organizations should provide IS employees with greater career opportunities, possibly by establishing dual career paths (managerial and technical career paths) to expand the career options within the IS department.

At the same time, the importance of salaries should not be underestimated. It was found to be negatively correlated with intention to voluntarily leave. Organizations need to be concerned about tangible rewards since the decision to remain with an organization is strongly related to financial status. Organizations need to make sure they reward the best performers by increasing their salaries. While it is acknowledged that other factors such as tenure and seniority also affect salaries, it is important to note that salary is often cited as a primary reason for career move. Pay was ranked highly among the most important motivators for IS employees.

Career opportunity was found to be positively correlated with the intention to leave the organization: Alternative job options can trigger intentions to leave. There are still factors within the control of organizations that can offset the constant allure of the external marketplace.

(5) *Perceived job characteristics.* Among IS personnel, job characteristics play a very important role in affecting the decision to stay. IS personnel, compared to others, have greater need for challenge and growth in their jobs and a strong preference for jobs that allow them freedom and creativity. Our study shows that opportunities for challenge and freedom within jobs correlate neg-

Table 6
Relationships between perceived job characteristics and intention to leave the organization ($N = 464$).

Pursue your idea	−0.27 [a]
Build a professional reputation	−0.24 [a]
Work with competent colleagues	−0.25 [a]
Work on technically challenging tasks	−0.34 [a]
Work on organizationally important projects	−0.18 [a]
Have freedom to be creative and original	−0.33 [a]
Be highly respected by your peers	−0.28 [a]
Work on professionally important projects	−0.30 [a]
Be highly respected by top management	−0.21 [a]
Have a great deal of power and influence on the job	−0.27 [a]
Work on projects leading to advancement	−0.26 [a]
Receive substantial annual salary increases	−0.21 [a]
Receive a promotion within the next year or two	−0.15 [a]
Task-based rewards	−0.37 [a]
Organizational-based rewards	−0.28 [a]

[a] $p \le 0.001$.

atively with the intention to leave the organization (see *Table 6*). In addition, both task-based and organizational-based rewards are negatively correlated with the intention to leave. However, correlation between the task-based rewards and the intention to leave is much stronger than between the organizational-based rewards and the intention to leave. Previous studies of technical professionals emphasize the importance of interesting and challenging work, freedom and recognition. Data shows that computer managers and project leaders are more concerned about the organizational-based rewards than programmers, and the programmers and software engineers are more interested in task-based rewards than other groups. Organizations should recognize this diversity in the IS population. Employees should be encouraged to discuss their needs and values with their supervisors regularly, developing more challenging assignments and setting meaningful career goals. Organizational reward systems and career path opportunities must be made flexible enough to accommodate a diverse IS workforce.

(6) *Satisfaction and commitment*. The five individual sources of career satisfaction (progress, promotion, pay, status, and projects), as well as the overall measures of both job satisfaction and career satisfaction, are negatively correlated with the intention to leave. The overall job satisfaction and career satisfaction are negatively correlated with the intention to leave. "Satisfaction with project" is relatively strong as a correlate to overall satisfaction. Again we see the need for organizations to be particularly aware of the kinds of projects that are assigned to IS employees, paying careful attention to challenge and interest. Organizations should continuously measure, monitor and control employee satisfaction in order to keep their IS employees.

Loyalty to the organization was found to be negatively correlated with the intention to leave. Thus, IS managers desiring to reduce turnover need to focus on enhancing their subordinates' organizational commitment. Accepted approaches for IS managers to consider include: Developing stronger co-worker relationships through team building; providing better supervision; improving reward systems; assigning more interesting and challenging projects; recognizing their efforts; providing more freedom and autonomy to pursue their ideas, and providing dual career paths.

4. Conclusions

Within this study we have identified significant factors affecting the intentions of IS personnel to voluntarily leave their organization. At the same time, we have provided a set of suggestions that could produce greater levels of job and career satisfaction. Specifically: Reducing role stressors such as role ambiguity and role conflict; encouraging boundary spanning activities and interactions with other internal and external organizational units; assigning challenging and interesting projects that allow for autonomy and creativity; providing ample recognition for achievement; and developing dual career ladders which allow for advancement within the IS field. There is little doubt that if managers in both the functional IS departments and larger organizational units implement these recommendations, the turnover intentions of IS employees could be significantly reduced.

Reviewing the current projections for growing shortages of new IS personnel and the increasing demands for them, as well as the previously mentioned economic costs associated with high turnovers, it would seem that most organizations should willingly incorporate these recommendations. Unfortunately, there are several forces at work today that could both delay this process as well as detract from its usefulness.

First, many of the recommendations involve organizational commitments that might not show significant results for several years. Studies show that serious attempts to change organizational climate and other intrinsic turnover correlates do not occur overnight. It is only when superficial approaches for solving complex human resource problems fail that the more time consuming and long term solutions are considered.

Second, meeting the individual needs of IS personnel may compromise higher level organizational goals. Conditions such as autonomy in decision making, interesting work, challenging jobs, and important projects, may reduce turnover, but could also detract from organizational effectiveness in maintaining rapid IS systems development cycles. McClelland's [11], work in the field of achievement motivation cautions that highly achievement oriented professionals, like IS employees, sometimes view the organization as existing primarily to serve their needs. For example, the 'perfect' software package or systems flow model might be too expensive for the customer or take longer then the desired delivery cycle. Or, the not so glamorous parts of any project might be postponed and avoided simply because they do not appear satisfying to the employee. To offset this problem, many firms employ professionals who not only remove uninteresting work from the professional's job, but it also allows the professional to spend a greater percentage of time engaged in greater income producing activities.

Third, most organizations large enough to support functional IS departments traditionally maintain a centralized human resource department with responsibility for all personnel. As such, there could be conflicts between the human resource policies and guidelines recommended for IS personnel and those possibly more suitable for other professional employees in the organization. Several large corporations have recognized this problem and created separate human resource functions within their IS departments.

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